Sankalp Jha

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Education

Year	Degree/Certificate	Institute	CPI/%
Oct 2022 – Aug 2026	B.Tech in CS & IT	Ajay Kumar Garg Engineering College, Delhi NCR	7.25/10
2021 – 2022	Senior Secondary (XII)	New Era School, Delhi NCR	91.2%
2017 – 2020	High School (X)	SJN International School, Coimbatore	92.6%

Experience

Contributor | *P4 Language Consortium (GSoC 2025), Stanford, CA* (*May 2025 – Sept 2025*)

Objective	To optimize packet inference and automate control-plane validations for the SpliDT framework.
Approach	• Engineered a Jinja2-based testing pipeline automating **25+** P4Runtime validations on Intel Tofino and Mininet. • Implemented Partitioned Decision Trees (PDTs) with TCAM-efficient Subtree IDs (SIDs) to resolve ASIC constraints.
Impact	Reduced packet inference latency by **20%** via optimized recirculation and improved deployment reliability by **30%**.

Key Projects / Open Source Contributions

Open Source Contributions (*Various*)

Overview	• **WasmEdge (PR #4470, #4466, #4440):** Patched undefined-behavior and memory-safety issues in the C++ host-function bindings and WASM execution engine; fixes were merged into the main runtime, improving spec-compliance and cross-platform stability. • **Facebook/bpfilter (PR #507):** Extended the eBPF-based packet-filtering daemon by adding missing rule-matching logic and hardening the netlink interface; the patch was accepted into the upstream codebase, strengthening kernel-level traffic filtering correctness.

SpliDT: Network Traffic Analysis with ML | *P4, Python, P4Runtime, Docker, Python, C++, P4Studio, Scripting* (*Project Link*)

Objective	To compile machine learning decision trees directly into P4 data planes using Partitioned Decision Trees (PDTs).
Approach	• Architected a switch-native ML framework overcoming ASIC memory limits via Subtree ID (SID) recirculation.
Impact	Achieved **10 Gbps** line-rate throughput and **5x** lower latency compared to standard control-plane inference.

P4 Lens | *React, FastAPI, Docker, React Flow* (*Project Link*)

Objective	To engineer a full-stack P4 visualizer that parses major language standards (P414/P416).
Approach	• Built interactive pipeline topologies rendering in under **200ms**, visualizing 4 core architectural blocks.
Impact	Reduced control-plane debugging time by **∼40%** through granular header and table inspection.

Stelltron | *Go, TypeScript, Rust, Python, C++* (*Project Link*)

Objective	A Decentralized Energy Grid on Stellar Ecosystem which turns energy into Tokenized Assets, and Energy into Stable Income.
Approach	• Creating Hardware Mesh Server of RaspberryPi3B and ESP32's pinging to our Software Server • Built a Robust Go Backend and 3 Smart Contracts on Soroban, with intuitive UI
Impact	Won Asia-Pacific across Bounties in Web3 space and recognized by Govt. Of India

Does My Idea Exist? | *TypeScript, Python, Dockerfile, JavaScipt, VectorDB, LLM/AI* (*Project Link*)

Objective	Built end-to-end SaaS Product which iterates over 500,000 Startups and help to answer whether their idea is unique?
Approach	• A cron-job scraper, which updates our Database Periodically and have consistently updated records • LLM inferencing the request of users and properly filtering via backend to relevant requests
Impact	Raised over USD10,000, and active product being used by over 100+ users across Globe

Technical Skills

- Python, C, C++, JavaScript, TypeScript, Go, P4
- **Libraries/Frameworks:** NodeJS, Numpy, Matplotlib, eBPF, gRPC, Protobufs, Next.js, Scapy, MLIR, LLVM, Sklearn, React, Web Services
- **Tools/Platforms:** Docker, CI/CD, VS Code, UTM, GitHub, GitLab, DaVinci Resolve, Linux (x86_64, ARM64 & AMD64)
- **Databases:** SQL, MongoDB, GraphQL, Redis, PostgreSQL, MySQL

Positions of Leadership

- **Technical Writer, Cloud Native Community Group, New Delhi** (*May 2024 – Dec 2024*)
  - Led a team of **6+** content writers and **4** designers to deliver documentation and event management support.
  - Partnered with marketing to revamp social media presence across LinkedIn, Instagram, and Twitter, achieving a **60%** increase in follower engagement.
- **Co-Organizer & Anchor, Google Developer Groups & GDSC WOW 2024, Delhi NCR** (*Nov 2023 – Nov 2025*)
  - Co-led **10+** tech events and anchored the GDSC WOW flagship event at IIIT Delhi, coordinating **31+** crew members across web, social, and on-ground operations.
  - Engaged a live audience of **800+** at WOW and **500+** developers overall, boosting participation by **30%** and visibility by **40%**.

Publications

- R. Bharadwaj, **S. Jha**, V. Malyan, R. Gupta, "Trusture: A Blockchain-Based Decentralized Framework for Secure, Transparent, and Auditable NGO Transactions," *2026 International Conference on Computing, Sciences and Communications (ICCSC)*, Ghaziabad, India, Feb. 2026. IEEE Xplore ↗ DOI: 10.1109/ICCSC67078.2026.11468597

Certifications & Honors

Certifications	• **Open Science 101** – NASA • **Data Science for Engineers** – NPTEL • **Enhancing Soft Skills & Personality** – NPTEL • **IIRS Satellite Remote Sensing** – ISRO
Honors & Awards	• **2x** AKTU State Men Singles TT Gold (2023, 2024) • International Robotics Olympiad Bronze, Thailand (2019) • Merit Certification at Carnic Model United Nation (2021) • NCC Cadet, National Cadet Corps (2019)

Living Portfolio Highlights

This appendix is synchronized from the projects and merged upstream contributions shown at sankalpjha.dev. Last generated: 2026-07-15.

Selected Projects

- **Los-Tecnicos** (TypeScript, go, iot, stellar): A Decentralized Energy Grid on Stellar Ecosystem which will turn energy into Tokenized Assets, and Energy into Stable Income. Repo: , a super cool residency project
- **DMIE**: A simple solution to every person's question, does my idea already exists, and also get your idea's comparative analysis of existing competition as we scraped and built on 500,000 Indian Startups. Private Repo in Collab with a Cracked DevFriend of mine.
- **Do-It** (Go, golang, local-first, multimedia): A self-hosted binary (for Raspberry Pi, adv. routers, old phones), built in Go, running SSE/HTTP server, allowing simple synced notes/multimedia files across all devices connected to ur local home wifi.
- **GSoC- SpliDT: Scaling Stateful Decision Tree Algorithms in P4** (HTML, gsoc, p4, p4lang): It is a switch-native compiler framework that enables stateful decision tree inference directly in programmable switches, bringing real-time machine learning into the network data plane. IT helps to compile high-performance decision tree models to enable detection and observability of security-significant flow behaviors across diverse traffic workloads on Intel Tofino ASICs. Report: (Codebase Private due to Research nature of the project)
- **p4Lens** (JavaScript): An Interactive P4 Program Visualiser, which allows to Upload and visualise P4 programs with an intuitive and interactive pipeline flow diagram, to those who are beginning to learn P4Lang.
- **Slix** (JavaScript): A full-stack movie streaming and review platform, built with React and Go, where admins can add movies and reviews, and Azure OpenAI does sentiment analysis on curator reviews so users can understand movie vibe at a glance. Live:

Selected Merged Pull Requests

- **WasmEdge/WasmEdge #4470**: fix(wasi): resolve 100% CPU usage on macOS due to clock mismatch (10,716 stars)
- **WasmEdge/WasmEdge #4440**: Upgrade Alpine base to 3.23 and restore static LLD support (10,716 stars)
- **WasmEdge/WasmEdge #4466**: feat(docker): add standalone alpine-base image build pipeline (10,716 stars)
- **WasmEdge/WasmEdge #4452**: docs(README): fix broken introduction and blog links (10,716 stars)
- **p4lang/tutorials #681**: Update README.md in reference to #680 (1,580 stars)
- **facebook/bpfilter #507**: style: add missing braces to satisfy readability-braces-around-statem... (343 stars)